Chapter-3 Mean Value Theorems

This chapter covers -

- Rolle's Theorem and corollaries - including CMVT, LMVT and L'Hopital's rule
- Convex and concave functions, Basic Slope Lemma, Points of inflection
- Taylor's theorem

That's just how I Rolle

Theorem (Rolle's Theorem)

Let $f : [a, b] \to \mathbf{R}$ *be a* **continuous** *and* **differentiable** *function. Also suppose* $f(a) = f(b)$.
Then there exists $c \in (a, b)$ *such that* $f'(c) = 0$.

Useful in proving statements regarding roots of a function.

Corollary 1

Suppose $f, g, h : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0$$

A Cauchy-ous approach to Mean Values

Theorem (Cauchy's Mean Value Theorem)

Suppose $f, g : [a, b] \to \mathbb{R}$ *are continuous on* $[a, b]$ *and differentiable on* (a, b). *Then there exists* $c \in (a, b)$ *such that*

$$f'(c)\left(g(b) - g(a)\right) \;=\; g'(c)\left(f(b) - f(a)\right).$$

It's a Secant? It's a Tangent!? It's LMVT!

Theorem (Lagrange's Mean Value Theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Sometimes useful in proving certain inequalities.

L'Hopital's got the cure

Theorem (L'Hopital's Rule)

Suppose $f, g : (a, b) \to \mathbb{R}$ *are differentiable and* $p \in (a, b)$ *such that* $g(x) \neq 0$ *and*

$$f(p) = g(p) = 0, \quad g'(x) \neq 0 \text{ on } (a, b).$$

Then

$$\lim_{x \to p} \frac{f(x)}{g(x)} = \lim_{x \to p} \frac{f'(x)}{g'(x)}$$

if that limit exists.

On roots and derivatives

Theorem

If c is a k-fold root of a polynomial $p(x)$, namely

$$p(x) = (x - c)^k Q(x), \quad Q(c) \neq 0,$$

then

$$p(c) = 0, \quad p'(c) = 0, \quad \ldots, \quad p^{(k-1)}(c) = 0 \tag{$*$}$$

and

$$p^{(k)}(c) \neq 0.$$

Converse: *If $(*)$ holds, then c is a k-fold root of $p(x)$.*

Exercise 1

$$f(x) = x + \frac{1}{x}, \qquad a = \frac{1}{2},\ b = 2$$

Verify LMVT by finding $c \in (a, b)$ that satisfies $f(b) - f(a) = f'(c)(b - a)$

Exercise 2

Let p and q be two real numbers with $p > 0$. Show that $x^3 + px + q$ has only one real root.

Exercise 3

Show that $f(x) = x^4 + 2x^3 - 2$ has exactly one root in (0,1).

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Solution -

$$f(1) = 1, \quad f(0) = -2$$

By IVP, at least one root exists in $(0, 1)$. Suppose two roots, a and b, exist in $(0, 1)$.

By Rolle's Theorem,

$$f'(c) = 0, \quad c \in (a, b)$$

Notice that $f'(x) = 4x^3 + 6x^2 = 0$ has no solutions in the interval $(0, 1)$.
Contradiction!!
Hence proved.

Exercise 4 - Inequality

Prove that $|\cos a - \cos b| \leq |a - b|$ for all $a, b \in \mathbf{R}$.

Exercise 4 - Inequality (Use LMVT ;))

Prove that $|\cos a - \cos b| \leq |a - b|$ for all $a, b \in \mathbf{R}$.

Solution -
Let

$$f(x) = \cos x$$

Apply LMVT in $[a, b]$:

$$f'(c) = -\sin c = \frac{\cos a - \cos b}{a - b}$$

Therefore,

$$|\cos a - \cos b| = |\sin c|\,|a - b| \leq |a - b|$$

Exercise 5 - Inequality again

Prove that

$$\frac{1}{2}\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots+\frac{1}{\sqrt{n}}\right)+1<\sqrt{n}<\frac{1}{2}\left(\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\ldots+\frac{1}{\sqrt{n-1}}\right)+1$$

for all $n \in \mathbb{N}$.

Exercise 5 - Inequality again (LMVT again)

Let $f(x) = \sqrt{x}$

Apply LMVT in $[n, n+1]$

$$\exists c_n \in (n, n+1) \text{ s.t. } f'(c_n) = \frac{1}{2\sqrt{c_n}} = \sqrt{n+1} - \sqrt{n}$$

$$\frac{1}{2\sqrt{n+1}} < \sqrt{n+1} - \sqrt{n} < \frac{1}{2\sqrt{n}}$$

$$\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{3}} + \cdots + \frac{1}{2\sqrt{n+1}} < \sqrt{n+1} - 1 < \frac{1}{2\sqrt{1}} + \frac{1}{2\sqrt{2}} + \cdots + \frac{1}{2\sqrt{n}}$$

$$\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{3}} + \cdots + \frac{1}{2\sqrt{n+1}} + 1 < \sqrt{n+1} < \frac{1}{2\sqrt{1}} + \frac{1}{2\sqrt{2}} + \cdots + \frac{1}{2\sqrt{n}} + 1$$

Hence proved.

Exercise 6

Prove that the polynomial

$$\left(\frac{d}{dx}\right)^n (x^2 - 1)^n$$

has n distinct roots in $(-1, 1)$.

Convex ~~lens~~ functions :o

A function $f : I \to \mathbb{R}$ defined on open interval I, is said to be **convex** if given $x, y \in I$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \tag{1}$$

for all $t \in [0, 1]$.
In other words, the chord joining $(x, f(x))$ and $(y, f(y))$ lies **above** the graph of f.

Identifying convex functions

Theorem

Let $f : I \to \mathbf{R}$ be twice-differentiable. Then -

$f'' \geq 0$ *on* I *iff* f *is convex on* I.

$f'' \leq 0$ *on* I *iff* f *is concave on* I.

Note that there can be convex/concave functions which are not twice-differentiable, in which case this theorem doesn't apply to such functions. However, our earlier definition still holds in such cases.

Basic Slope Lemma

Let $f : I \to \mathbb{R}$ be convex.
Assume $x < y < z$; $x, y, z \in I$.
Then the slope of the chord joining $(x, f(x))$ and $(y, f(y))$ satisfies

$$\frac{f(y) - f(x)}{y - x} \quad \leq \quad \frac{f(z) - f(y)}{z - y},$$

i.e.

$$\text{Slope } (x, y) \quad \leq \quad \text{Slope } (y, z)$$

Exercise 7

Prove the basic slope lemma.

Exercise 7 - Solution

Prove the basic slope lemma.
Solution - From the definition of convex functions,

$$f(y) \leq tf(x) + (1-t)f(z),$$

We get using the section formula,

$$t = \frac{z-y}{z-x}, \quad 1-t = \frac{y-x}{z-x}.$$

$$\therefore \quad (z-x)f(y) \leq (z-y)f(x) + (y-x)f(z).$$

Rearranging, you get

$$(z-y)\big(f(y)-f(x)\big) \leq (y-x)\big(f(z)-f(y)\big).$$

$$\therefore \quad \frac{f(y)-f(x)}{y-x} \leq \frac{f(z)-f(y)}{z-y}.$$

Hence proved

A Matter of Inflection

The point $x = c$ is called a **point of inflection** of $f(x)$, if in a small interval $(c - \delta, c)$, $f(x)$ is convex; and in a small interval $(c, c + \delta)$, $f(x)$ is concave, or vice-versa.
Basically, at a point of inflection, the function changes from being convex to concave or vice-versa. The graph crosses its tangent and bends away from the tangent in the other direction.

Theorem (Point of inflection)

If $f(x)$ is twice-differentiable at $x = c$, and $x = c$ is a point of inflection of $f(x)$, then $f''(c) = 0$.

Taylor-made for you ♡

If $f : I \to \mathbb{R}$ is n-times differentiable and $a, x \in I$.
Then, $\exists\, c$ between a and x such that

$$f(x) = f(a) + (x-a)f'(a) + \cdots + \frac{(x-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n(a,x,c)$$

where

$$R_n(a,x,c) = \frac{(x-a)^n}{n!}f^{(n)}(c).$$

$R_n(a,x,c)$ is called the Young's form of remainder.

Seriesly useful formulas

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots,$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \qquad |x| < 1.$$

Exercise 8

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^3 + 3^2, & x \leq 0, \\ (x-3)^2, & x > 0, \end{cases}$$

Which of the following are true?

(a) The function $f(x)$ is strictly convex in the interval $(1, 5)$.

(b) The function $f(x)$ is differentiable at all points in $\mathbb{R}$.

(c) The point $x = 0$ is a point of inflection for the function $f(x)$.

(d) The function $f(x)$ has a local maximum at $x = 0$.

Exercise 8

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(c) The point $x = 0$ is a point of inflection for the function $f(x)$.

(d) The function $f(x)$ has a local maximum at $x = 0$.

Answer - (a), (c)

Exercise 9

Prove that for $x \in (0, 1)$,

$$e^x < 1 + (e - 1)x$$

Exercise 9 - Solution

Prove that for $x \in (0, 1)$,

$$e^x < 1 + (e - 1)x$$

Hint - Use the fact that e^x is a convex function. Then apply the definition of convex functions in the interval $[0, 1]$.

Exercise 10

If f and g have a point of inflection at $x = 0$, then fg has a point of inflection at $x = 0$. True/False?

Exercise 10 - Solution

If f and g have a point of inflection at $x = 0$, then fg has a point of inflection at $x = 0$. True/False? Let us try to find a counter-example. Let

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$$f(x) = \begin{cases} x^2, & x < 0 \\ -x^2, & x > 0 \end{cases}$$

$$g(x) = \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases}$$

$$(fg)(x) = -x^4$$

$x = 0$ is not a point of inflection. Hence, the answer is **False**.

Exercise 11

A function is continuously differentiable if it is differentiable, and its derivative is continuous. Which one of the following statements is **FALSE**?

For any continuously differentiable function $g : [0, 1] \to \mathbb{R}$ with

$$g(0) = 1, \quad g(1) = e, \quad \text{and} \quad g'_+(0) = \frac{1}{2},$$

there is $c \in (0, 1)$ such that

(a) g'(c) = 2c g(c)

(b) g'(c) = 1

(c) g'(c) = e - 1

(d) g'(c) = 2e

Exercise 11 - Solution

Proof of part (c)

Apply LMVT to $g(x)$ in $[0, 1]$

$$\therefore \exists c_1 \text{ s.t. } g'(c_1) = \frac{g(1) - g(0)}{1 - 0} = e - 1$$

Proof of part (b)

Since derivative is continuous,
Apply IVP to $g'(x)$

$$g'(0^+) = 0.5$$

$$g'(1^-) = e - 1 \approx 1.7$$

$$\therefore \exists c_2 \in (0, c_1) \text{ s.t. } g'(c_2) = 1$$

Proof of part (a)

Define

$$f(x) = e^{-x^2} g(x)$$

$$f(0) = f(1) = 1$$

Apply Rolle's Thm in $[0, 1]$

$$\therefore \exists c_3 \in (0, 1) \text{ s.t. } f'(c_3) = 0$$

$$\implies e^{-c_3^2}(g'(c_3) - 2c_3 g(c_3)) = 0$$

$$\therefore g'(c_3) = 2c_3 g(c_3)$$

Exercise 11 - Solution

A function is continuously differentiable if it is differentiable, and its derivative is continuous. Which one of the following statements is **FALSE**?

For any continuously differentiable function $g : [0, 1] \to \mathbb{R}$ with

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there is $c \in (0, 1)$ such that

(a) g'(c) = 2c g(c)

(b) g'(c) = 1

(c) g'(c) = e - 1

(d) g'(c) = 2e

Answer - (d)

Can we go home now?

Not so fast :)

Don't stress too much about grades. How many marks you score in this exam will not affect the number of zeroes in your salary.
So,

(Credits for this slide - Dhairya and Niral from last year's TSC :D)

All the best!